

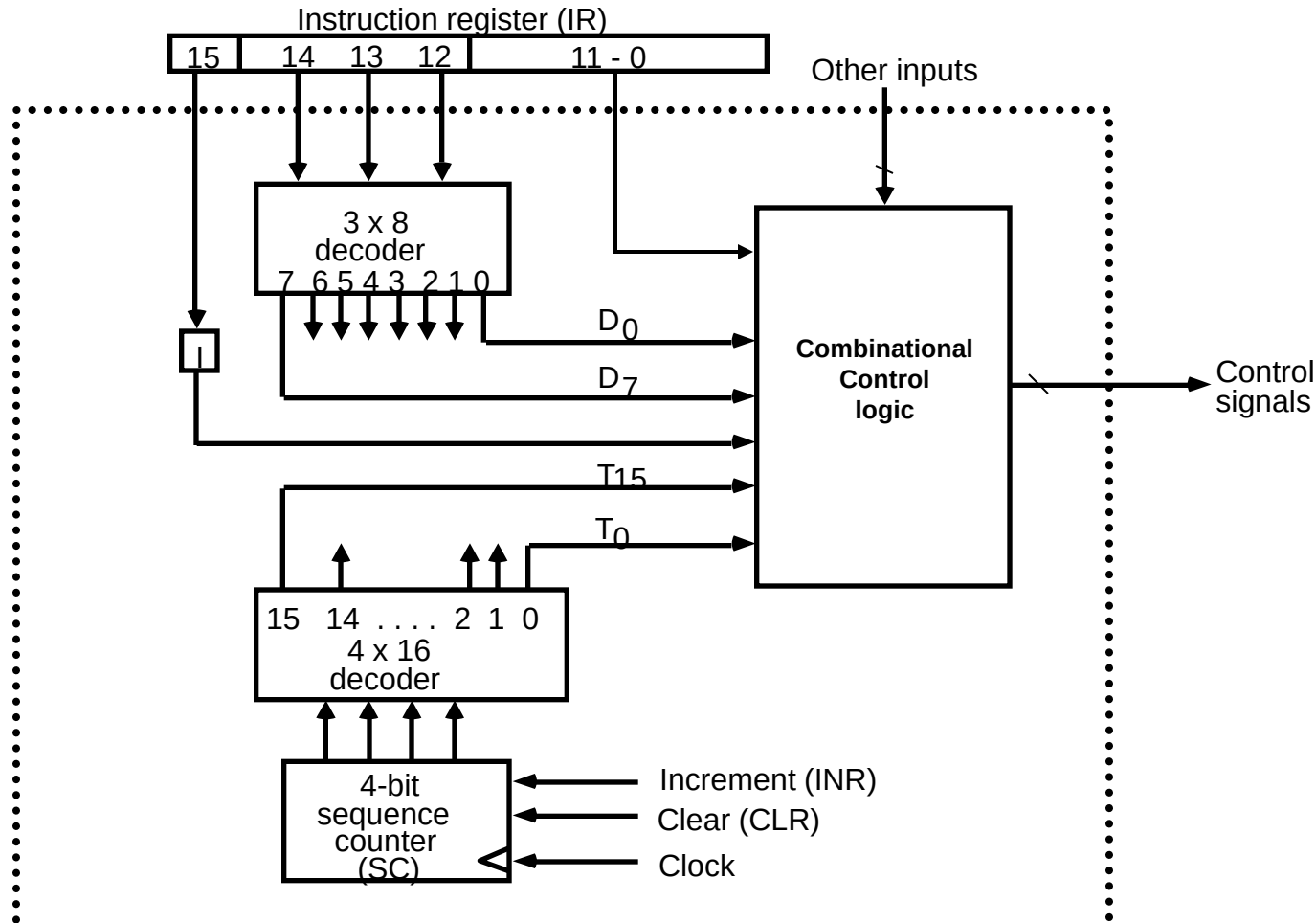
CONTROL UNIT

- Control unit (CU) of a processor translates machine instructions into control signals for the microoperations that implement them.
- Control units are implemented in one of two ways
- *Hardwired Control*
 - CU is made up of sequential and combinational circuits to generate the control signals
- *Microprogrammed Control*
 - A control memory on the processor contains microprograms that activate the necessary control signals
- We will consider a hardwired implementation of the control unit for the Basic Computer



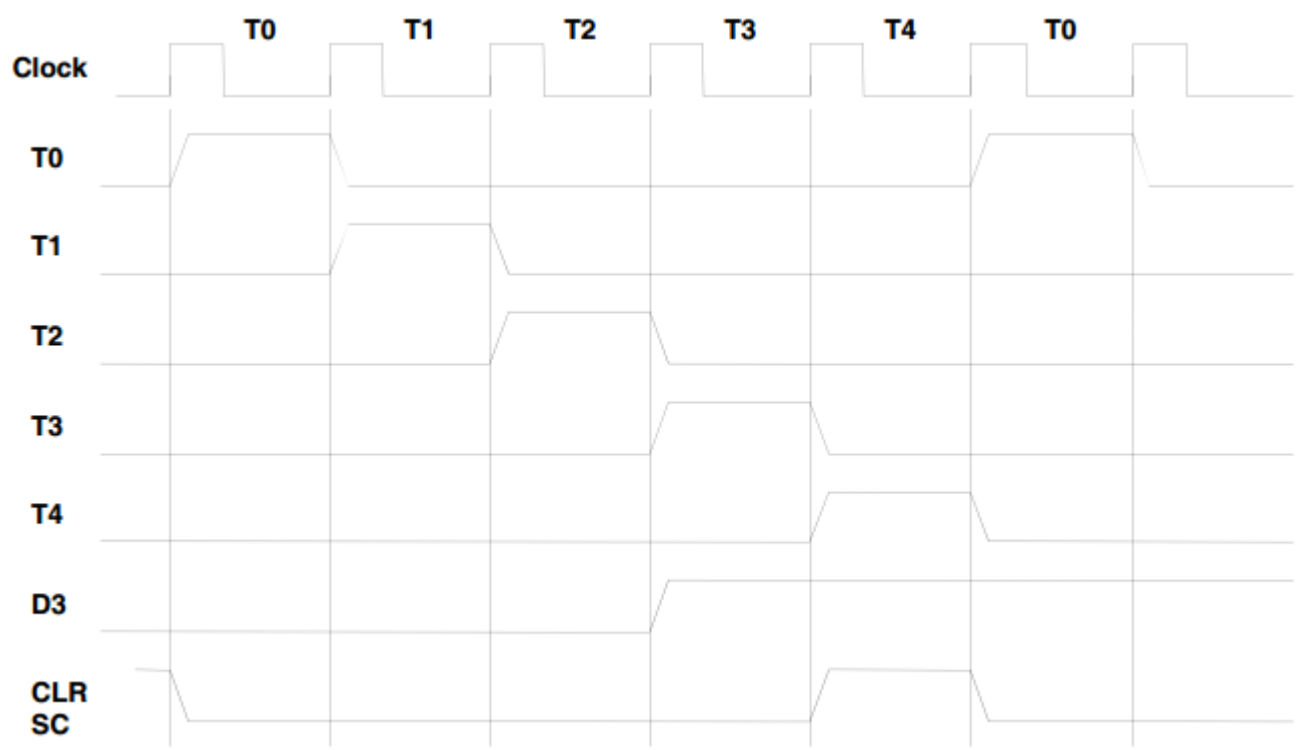
TIMING AND CONTROL

Control unit of Basic Computer



TIMING SIGNALS

- Generated by 4-bit sequence counter and 4×16 decoder
- The SC can be incremented or cleared.
- Example: $T_0, T_1, T_2, T_3, T_4, T_0, T_1, \dots$
Assume: At time T_4 , SC is cleared to 0 if decoder output D3 is active.
 $D_3 T_4: SC \leftarrow 0$



INSTRUCTION CYCLE

- In Basic Computer, a machine instruction is executed in the following cycle:
 1. Fetch an instruction from memory
 2. Decode the instruction
 3. Read the effective address from memory if the instruction has an indirect address
 4. Execute the instruction
- After an instruction is executed, the cycle starts again at step 1, for the next instruction
- *Note:* Every different processor has its own (different) instruction cycle



Memory Read/Write

- Normally in a modern processors, the processor is trying to complete one basic part of instruction per clock cycle.
- Usually memory read/write cycle consumes several clk pulses.
- If processor requires some data from memory for currently executing instruction than it cant do anything except wait.
- It will be assumed that a memory cycle time is less than the clock cycle time.



FETCH and DECODE

- Fetch and Decode

T0: $AR \leftarrow PC$ ($S_0S_1S_2=010$, $T_0=1$)

T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC + 1$ ($S_0S_1S_2=111$, $T_1=1$)

T2: $D_0, \dots, D_7 \leftarrow \text{Decode } IR(12-14)$, $AR \leftarrow IR(0-11)$, $I \leftarrow IR(15)$

